

What the exponential integrals are telling you

Why Ei blows up at $\Lambda \rightarrow 0$ and at $y' \rightarrow z$, and which of the two is a divergence

$$\kappa \equiv k \cdot (y' - z), \quad \lambda \equiv \Lambda/k^+, \quad \Xi \equiv (\vee - k^+)/k^+, \quad \ell_k \equiv \log \frac{k^+}{\Lambda}$$

1. Why it blows up in both limits: Ei only sees the product

Your p^+ integral produces $\text{Ei}(-i\kappa\lambda)$, and for small argument

$$\text{Ei}(-iu) = \gamma_E + \log(iu) + O(u).$$

The argument is the **product** $u = \kappa\lambda$. Mathematica cannot tell you whether it became small because $\Lambda \rightarrow 0$ or because $y' \rightarrow z$ — it is the same number:

kappa=1	lam=1e-06	product=1e-06:	Ei = -13.23829489 - 1.570797j
kappa=0.001	lam=0.001	product=1e-06:	Ei = -13.23829489 - 1.570797j
kappa=1e-06	lam=1	product=1e-06:	Ei = -13.23829489 - 1.570797j
kappa=2	lam=5e-07	product=1e-06:	Ei = -13.23829489 - 1.570797j

So the blow-up is one logarithm, which splits as

$$\log(i\kappa\lambda) = \underbrace{\log \lambda}_{=-\ell_k} + \underbrace{\log(i k \cdot (y' - z))}_{\text{a function of the transverse variables}}.$$

These two are completely different objects. The first is the rapidity divergence — real, and the thing you subtract. The second is a logarithmic *singularity of the integrand at a point*, which is a different animal from a divergence of the integral.

2. The $y' \rightarrow z$ piece is *not* a transverse divergence

A logarithm is integrable in two dimensions. $\log(y' - z)^2$ is also “undefined” at $y' = z$, yet $\int d^2(y' - z) \log(y' - z)^2$ is perfectly finite. Here the integrand near $y' = z$ is $\frac{1}{\rho} \times \log \rho$ and

$$\int_0 \rho d\rho \frac{\log \rho}{\rho} = \int_0 d\rho \log \rho < \infty.$$

Measured — the disc integral of the p^+ -integrated integrand, the one with the Ei's in it:

int_{ y'-z < 1e-01} d^2y'	= 9.2229e-04	
int_{ y'-z < 1e-02} d^2y'	= 1.0765e-05	ratio 0.0117
int_{ y'-z < 1e-03} d^2y'	= 1.0785e-07	ratio 0.0100

0.01 per decade means $F \propto \epsilon^2$ — it does better than merely converging, because the leading $\frac{r^m}{r^2} \log |r|$ still averages to zero over angles.

So: no. The Ei being undefined at $y' = z$ does not mean a transverse divergence, and it does not obstruct the + prescription. It is the rapidity divergence wearing a transverse disguise, because Ei reports only $\kappa\lambda$.

3. But the same logarithm does bite — at the other end

Take $\vee \rightarrow \infty$ properly. $\text{Ei}(-i\kappa\Xi) \rightarrow -i\pi \text{sgn } \kappa$ (a constant, not zero), so the bracket becomes

$$-[\text{Ei}(-i\kappa\Xi) - \text{Ei}(-i\kappa\lambda)] \longrightarrow -\ell_k + \gamma_E + \log(i k \cdot (y' - z)),$$

which **grows like** $\log |y' - z|$ at large separation, while the kernel structure falls only as $1/r^2$ with a nonvanishing angular average $(\rho^2 \langle T_2 \rangle) = +0.028221$, constant from $\rho = 10$ to 10^3 , matching $-\frac{d_e}{2d^2 \epsilon^2}$. So

$$\int^R d^2r \frac{1}{r^2} \left[-\ell_k + \log |r| \right] \sim -c \ell_k \log R + \frac{c}{2} \log^2 R, \quad c = 2\pi \langle T_2 \rangle \rho^2.$$

The truncated integral confirms it, and quantitatively:

R	$F(R)$	$dF/d\log R$	change per decade
10^2	-6.48980	—	—
10^3	-10.05680	-1.5491	—
10^4	-12.68358	-1.1408	+0.4083
10^5	-14.37029	-0.7325	+0.4083
10^6	-15.12141	-0.3262	+0.4063

The slope does not settle — it **grows linearly in $\log R$** , by a constant 0.4083 per decade. That is the signature of $\frac{c}{2} \log^2 R$, and the rate is predicted exactly:

$$c = 2\pi \times 0.028221 = 0.17732, \quad c \log 10 = 0.4083.$$

Both pieces are divergent: $-c \ell_k \log R$ is the transverse divergence multiplying the rapidity log, and $\frac{c}{2} \log^2 R$ is the transverse divergence squared. The second is precisely where the ℓ_k^2 comes from once R is cut by $1/(\xi|k|)$.

4. Summary, and one Mathematica caution

question	answer
Why is Ei undefined at $\Lambda \rightarrow 0$?	rapidity divergence, $\log \lambda = -\ell_k$ — real
Why also at $y' \rightarrow z$?	Ei sees only $\kappa \lambda$; it is the same log
Does $y' \rightarrow z$ mean a transverse divergence?	No — a log integrates fine in 2D; $F \propto \epsilon^2$
Can you do the + prescription?	Yes , as an operation; it is an exact identity
Are the two resulting terms separately finite?	No — large $ y' - z $, §3

Branch caution. ExpIntegralEi of an imaginary argument sits on a branch cut. The relation to the principal-branch E_1 is

$$\text{Ei}(-iu) = -E_1(iu) - i\pi \operatorname{sgn}(u) \quad (u \text{ real}),$$

checked to 10^{-10} . In your difference the $-i\pi \operatorname{sgn}$ cancels, since Ξ and λ are both positive — so the bracket is safe. But do not drop $\text{Ei}(-i\kappa\Xi)$ when taking $\vee \rightarrow \infty$: it tends to the constant $-i\pi \operatorname{sgn} \kappa$, not to zero, and that constant is a genuine absorptive contribution.

zint/src/ei_limits.py, zint/src/ei_slope.py — every number above, written to zint/RESULTS_ei.txt.